

D787

Scalability and Performance Optimization

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A stateless web API runs on one server that hits 90% CPU at peak. It is already on the largest instance size the cloud provider offers. The marketing team just redesigned the product logo. What is the best way to add capacity?

- A. Nothing; scaling is impossible once you reach the largest instance.
- B. Scale up only the database.
- C. Scale vertically to a larger instance.
- D. Scale horizontally: add more servers behind a load balancer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A batch job takes 100 minutes on one core, and 80% of its work can run in parallel. The team moves it to 4 cores. The job was written in 2019 by an intern. What is the maximum speedup?

- A. 3.2 times
- B. 5 times
- C. 4 times
- D. 2.5 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A program is 90% parallelizable. With unlimited processors, what is the most it can be sped up?

- A. 10 times
- B. Without limit, since processors are unlimited.
- C. About 1.11 times
- D. 90 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A checkout service receives 200 requests per second, and each request spends 0.5 seconds in the system on average. The service is written in Node.js and exposes 12 endpoints. On average, how many requests are in the system at once?

- A. 100
- B. 400
- C. 200
- D. 2,400

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A queue holds 60 requests on average, and requests arrive at 30 per second. What is the average time a request spends in the system?

- A. 1,800 seconds
- B. 2 seconds
- C. 0.5 seconds
- D. 60 seconds

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An API's average response time is 180 ms, under its 200 ms target, but its p99 is 2.4 seconds and users complain. It handles 1 million calls a day from a data center in Ohio. Which reading is correct?

- A. 99% of calls take 2.4 seconds.
- B. About 10,000 calls a day take 2.4 seconds or longer; the average hides this tail.
- C. Performance is fine because the average beats the target.
- D. Moving out of the Ohio data center will fix the p99.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Before a holiday sale, a team wants to confirm the site gives acceptable response times at its expected peak of 5,000 concurrent users. The team just finished moving its logs to a new tool. Which test fits?

- A. A stress test to find the breaking point.
- B. A soak test running for several days.
- C. A load test at 5,000 concurrent users.
- D. A single-user functional test of checkout.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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8.

A data pipeline has three stages: parse at 500 records per second, enrich at 120, and write at 300. The enrich stage came from a vendor with a strong reputation. What is the pipeline's maximum throughput, and where should tuning start?

- A. About 307 records per second, the average of the stages.
- B. 120 records per second; tune parsing since it runs first.
- C. 120 records per second; tune the enrich stage.
- D. 500 records per second; tune the write stage.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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9.

A cache hit takes 2 ms and a miss, which goes to the database, takes 50 ms. The hit ratio is 90%. The cache runs on 3 nodes. What is the average latency?

- A. 26 ms
- B. 20.4 ms
- C. 6.8 ms
- D. 45.2 ms

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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10. [TRANSFER — reversed stem / different variable]

A cache hit takes 1 ms and a miss takes 41 ms. What hit ratio gives an average latency of 5 ms?

- A.** 10%
- B.** 90%
- C.** About 12.2%
- D.** About 87.8%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A service fails on average every 190 hours (MTBF) and takes 10 hours to repair (MTTR). The team has an on-call rotation of 5 engineers. What is its availability?

- A.** 95%
- B.** 5%
- C.** It cannot be found without the on-call roster.
- D.** About 94.7%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Two independent servers each have 95% availability, and either one alone can serve all traffic. What is the combined availability?

- A.** 100%, since redundancy removes all downtime.
- B.** 90.25%
- C.** 95%
- D.** 99.75%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A stateless web API runs on one server that hits 90% CPU at peak. It is already on the largest instance size the cloud provider offers. ~~The marketing team just redesigned the product logo.~~ What is the best way to add capacity?

Target word: scaling approach **Decides:** No bigger instance exists; the app is stateless.

A	Absolute language	That is the limit of vertical scaling only.
B	Adjacent category	The CPU bottleneck is on the API server.
C	Confusion set	There is no larger instance available.
D	Correct	Stateless services scale out easily when vertical scaling is maxed.

2. Answer D Concept 2

A batch job takes 100 minutes on one core, and 80% of its work can run in parallel. The team moves it to 4 cores. ~~The job was written in 2019 by an intern.~~ What is the maximum speedup?

Target word: speedup on 4 cores **Decides:** Serial part is 20%.

A	Inverted operation	Multiplies 0.8×4 instead of applying the formula.
B	Confusion set	That is the limit with unlimited cores, $1 / 0.2$.
C	Dropped element	Ignores the 20% serial part.
D	Correct	$1 / (0.2 + 0.8 / 4) = 1 / 0.4 = 2.5$.

3. Answer A Concept 2 · transfer

A program is 90% parallelizable. With unlimited processors, what is the most it can be sped up?

Target word: upper limit of speedup **Decides:** Limit = $1 / \text{serial fraction}$.

A	Correct	$1 / 0.1 = 10$.
B	Absolute language	The 10% serial part never shrinks.
C	Inverted operation	Divides by the parallel fraction instead of the serial one.
D	Wrong stem number	Treats the 90 in the stem as the multiplier.

4. Answer A Concept 3

A checkout service receives 200 requests per second, and each request spends 0.5 seconds in the system on average. ~~The service is written in Node.js and exposes 12 endpoints.~~ On average, how many requests are in the system at once?

Target word: concurrent requests **Decides:** $L = \lambda \times W$.

A	Correct	$200 \times 0.5 = 100$.
B	Inverted operation	Divides 200 by 0.5 instead of multiplying.
C	Wrong stem number	The arrival rate, not the number in the system.
D	Camouflage	Multiplies by the endpoint count, which does not matter.

5. Answer B Concept 3 · transfer

A queue holds 60 requests on average, and requests arrive at 30 per second. What is the average time a request spends in the system?

Target word: solve for W **Decides:** $W = L / \lambda$.

A	Inverted operation	Multiplies instead of dividing.
B	Correct	$60 / 30 = 2$.
C	Confusion set	Flips the ratio: $30 / 60$.
D	Wrong stem number	The queue length, not the time.

6. Answer B Concept 4

An API's average response time is 180 ms, under its 200 ms target, but its p99 is 2.4 seconds and users complain. It handles 1 million calls a day ~~from a data center in Ohio~~. Which reading is correct?

Target word: interpret p99 **Decides:** p99 describes the slowest 1% of requests.

A	Inverted operation	p99 means 99% are faster than 2.4 seconds.
B	Correct	1% of 1,000,000 = 10,000 slow calls; targets should use percentiles.
C	Dropped element	Ignores the tail the complaints come from.
D	Camouflage	Location is not shown to cause the tail.

7. Answer C Concept 5

Before a holiday sale, a team wants to confirm the site gives acceptable response times at its expected peak of 5,000 concurrent users. ~~The team just finished moving its logs to a new tool.~~ Which test fits?

Target word: test type **Decides:** Goal is behavior at expected peak.

A	Confusion set	Stress tests push beyond expected load.
B	Adjacent category	Soak tests look for leaks over time, not peak response.
C	Correct	Load tests check performance at expected levels.
D	Dropped element	Drops the concurrent load entirely.

8. Answer C Concept 6

A data pipeline has three stages: parse at 500 records per second, enrich at 120, and write at 300. ~~The enrich stage came from a vendor with a strong reputation.~~ What is the pipeline's maximum throughput, and where should tuning start?

Target word: throughput and tuning target **Decides:** Slowest stage sets throughput.

A	Confusion set	Stages in sequence do not average.
B	Adjacent category	Speeding a non-bottleneck stage adds nothing.
C	Correct	The pipeline can go no faster than its slowest stage.
D	Wrong stem number	Uses the fastest stage's rate.

9. Answer C Concept 7

A cache hit takes 2 ms and a miss, which goes to the database, takes 50 ms. The hit ratio is 90%. ~~The cache runs on 3 nodes.~~ What is the average latency?

Target word: weighted average latency **Decides:** $0.9 \times 2 + 0.1 \times 50$.

A	Dropped element	Simple average that drops the hit ratio weights.
B	Camouflage	Multiplies by the node count, which does not matter.
C	Correct	$1.8 + 5.0 = 6.8$ ms.
D	Inverted operation	Swaps the weights: $0.1 \times 2 + 0.9 \times 50$.

10. Answer B Concept 7 · transfer

A cache hit takes 1 ms and a miss takes 41 ms. What hit ratio gives an average latency of 5 ms?

Target word: solve for hit ratio **Decides:** $h \times 1 + (1 - h) \times 41 = 5$.

A	Inverted operation	That is the miss ratio.
B	Correct	$41 - 40h = 5$, so $h = 0.9$.
C	Confusion set	$5 / 41$ compares the target to miss time, not a ratio of hits.
D	Dropped element	$(41 - 5) / 41$ drops the 1 ms hit cost.

11. Answer A Concept 8

A service fails on average every 190 hours (MTBF) and takes 10 hours to repair (MTTR). ~~The team has an on-call rotation of 5 engineers.~~ What is its availability?

Target word: availability **Decides:** $MTBF / (MTBF + MTTR)$.

A	Correct	$190 / (190 + 10) = 0.95$.
B	Inverted operation	That is the unavailability.
C	Camouflage	MTBF and MTTR already decide it.
D	Dropped element	$1 - 10/190$ drops repair time from the total period.

12. Answer D Concept 8 · transfer

Two independent servers each have 95% availability, and either one alone can serve all traffic. What is the combined availability?

Target word: parallel availability **Decides:** Both must fail for an outage.

A	Absolute language	Both can still fail at once.
B	Inverted operation	Multiplies as if both were needed (series).
C	Dropped element	Ignores the second server.
D	Correct	$1 - 0.05 \times 0.05 = 0.9975$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Vertical vs. horizontal scaling	Vertical scaling makes one machine bigger and hits a ceiling; horizontal scaling adds machines and suits stateless services.	Believing scaling stops at the largest machine.	A maxed-out instance and a stateless app.
2	Amdahl's law	$\text{Speedup} = 1 / ((1 - p) + p / n)$; the serial part caps the gain no matter how many processors.	Expecting n processors to give n times the speed.	A parallel fraction and a core count.
3	Little's law	Items in the system = arrival rate $\times$ average time in the system ($L = \lambda W$).	Dividing instead of multiplying.	A rate per second and an average duration.
4	Tail latency and percentiles	Averages hide slow requests; p95/p99 show what the slowest share of users experience.	Declaring success because the mean meets the target.	A good average next to a bad p99.
5	Types of performance tests	Load tests check expected peak; stress tests push past limits to find the breaking point; soak tests run long to find leaks; spike tests check sudden jumps.	Calling every performance test a stress test.	Whether the goal is expected load, breaking point, duration, or suddenness.
6	Bottlenecks	A pipeline's throughput equals its slowest stage; optimizing elsewhere does not raise it.	Averaging stage speeds or tuning the fastest stage.	Stage-by-stage throughput figures.
7	Cache effectiveness	Average latency = hit ratio $\times$ hit time + miss ratio $\times$ miss time.	Averaging hit and miss times without weights.	A hit ratio with hit and miss times.
8	Availability and redundancy	Availability = $\text{MTBF} / (\text{MTBF} + \text{MTTR})$; parallel redundancy gives $1 - (1 - a)^n$.	Leaving repair time out, or multiplying parallel parts as if in series.	MTBF and MTTR, or redundant components.

Course scope (public description): This course prepares students to learn how to analyze real-world systems to make informed decisions to improve overall design, performance, and reliability. Students will create tests that validate both scalability and reliability as well as apply theoretical to real-world design, performance, and reliability systems.